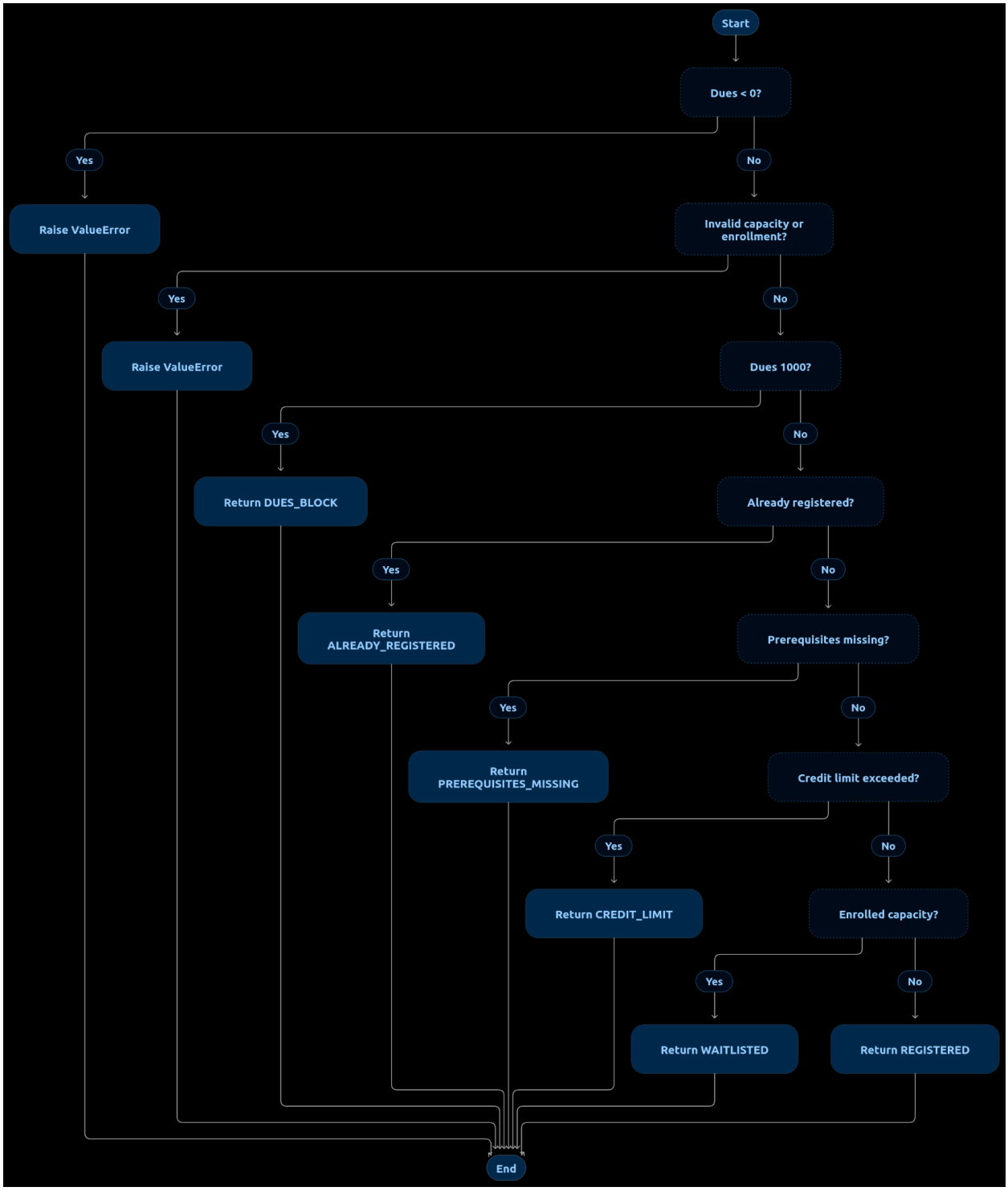
B1. Control Flow Graph (CFG)

A Control Flow Graph represents the execution flow of a function. It shows:

- Nodes: Individual statements or groups of statements.
- Edges: Possible paths of execution between nodes.
- Decision nodes: Conditions such as `if`, `else`, and logical expressions.


Start
Dues < 0?
Yes
No
Raise ValueError
Invalid capacity or enrollment?
Yes
No
Raise ValueError
Dues 1000?
Yes
No
Return DUES_BLOCK
Already registered?
Yes
No
Return ALREADY_REGISTERED
Prerequisites missing?
Yes
No
Return PREREQUISITES_MISSING
Credit limit exceeded?
Yes
No
Return CREDIT_LIMIT
Enrolled capacity?
Yes
No
Return WAITLISTED
Return REGISTERED
End

Calculate Cyclomatic Complexity

Cyclomatic Complexity (CC) measures the number of independent paths through a function.

The formula is:

$$CC = E - N + 2P$$

Where:

- E = Number of edges
- N = Number of nodes
- P = Number of connected components (normally 1 for one function)

For this CFG:

- Nodes $N = 17$
- Edges $E = 23$
- Connected components $P = 1$

$$CC = 23 - 17 + 2(1)$$
CC = 8

We can also calculate it using the simpler formula:

$$CC = Numberofdecisionpoints + 1$$

There are 7 `if` decision points in this function.

CC=7+1= 8

Meaning: We need at least 8 independent execution paths to cover the basic decision structure of this function. This does not mean we need only 8 test cases for every possible input combination.